

Interactive Expense Dashboard using Excel & VBA

1. Project Overview

The Interactive Expense Dashboard is a financial reporting solution developed using Microsoft Excel and VBA. It enables users to monitor expenses, compare budgets with actual spending, analyze departmental performance, and track monthly expense trends through an interactive dashboard.

This project demonstrates the application of Excel, VBA automation, Pivot Tables, Pivot Charts, and AI-assisted development to create a user-friendly financial analysis tool.

2. Objectives

- Develop an interactive expense dashboard.
 - Monitor business expenses efficiently.
 - Compare Budget vs Actual spending.
 - Analyze departmental expenses.
 - Visualize monthly expense trends.
 - Improve reporting through VBA automation.
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3. Tools & Technologies

- Microsoft Excel
 - VBA (Visual Basic for Applications)
 - Pivot Tables
 - Pivot Charts
 - Slicers
 - Claude AI (AI-assisted VBA development)
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4. Dashboard Features

- KPI Cards
 - Total Expenses
 - Budget Allocated
 - Budget Remaining
 - Transactions
 - Approval Rate
 - Average Expense
 - Department-wise Expense Analysis
 - Expense Category Analysis
 - Monthly Expense Trend
 - Budget vs Actual Comparison
 - Payment Method Analysis
 - Interactive Filters (Month & Department)
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5. VBA Automation

The project uses VBA to automate dashboard functionality, making the reporting process faster and more interactive.

Examples include:

- Dashboard refresh
 - Navigation automation
 - Dynamic reporting
 - Interactive controls
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6. AI-Assisted Development

Claude AI was used as a coding assistant to generate and refine portions of the VBA code through prompt engineering.

The dashboard design, customization, testing, validation, and financial analysis were completed by the project author.

7. Business Benefits

- Faster financial reporting
 - Better expense monitoring
 - Improved budget control
 - Interactive decision support
 - Reduced manual reporting effort
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8. Learning Outcomes

Through this project, I improved my skills in:

- Advanced Microsoft Excel
 - VBA Programming
 - Dashboard Design
 - Financial Reporting
 - Data Visualization
 - Prompt Engineering
 - AI-assisted Development
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9. Future Enhancements

- Power BI integration

- Python-based analytics
 - AI expense forecasting
 - Automated PDF reporting
 - Cloud database integration
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10. Project Files

- Expense_Dashboard.xlsm
 - Dashboard Screenshots
 - VBA Code
 - AI Prompt Documentation
 - README.md
 - Project Documentation
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11. Author

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